

Matrix

A matrix is a rectangular arrangement of numbers, symbols, or expressions organized in rows and columns. Each value is called an element, and its position is determined by its row and column.

Matrices are used to represent and organize data in a structured form. They are especially useful in solving systems of linear equations quickly and efficiently. In addition, matrices are widely applied in computer graphics, engineering, physics, and data analysis. Operations such as addition, multiplication, and finding determinants make matrices powerful tools for simplifying complex mathematical problems and performing large-scale calculations in various real-world applications.

Some mathematical operations of matrices are given below.

```
>> a = [1 2 3; 5 6 2; 1 0 4]
```

a =

```
1    2    3
5    6    2
1    0    4
```

```
>> b = [2 7 4; 3 0 1; 3 2 9]
```

b =

```
2    7    4
3    0    1
3    2    9
```

```
>> a + b % additon matrix a and b
```

ans =

```
3    9    7
8    6    3
4    2   13
```

```
>> a - b % subtraction matrix a and b
```

ans =

```
-1   -5   -1
2     6     1
-2   -2   -5
```

```
>> a * b % multiplication matrix a and b
```

ans =

```
17   13   33
34   39   44
14   15   40
```

```
>> a.*b % Element-wise multiplication
```

ans =

```
2    14   12
15    0    2
3     0   36
```

```
>> 2*b % Scalar multiplication
```

ans =

```
4    14    8
6     0    2
6     4   18
```

```
>> c = a' %transpose of a
```

c =

```
1    5    1
2    6    0
3    2    4
```

```
>> C = A.' % Conjugate transpose
```

C =

```
1
2
3
4
```

```
>> d = det(A) % Determinant
Error using det
Matrix must be square.
```

```
>> d = det(a) % Determinant
```

d =

```
-30
```

```
>> d = inv(a) % inverse
```

```
d =
```

```
-0.8000    0.2667    0.4667  
0.6000   -0.0333   -0.4333  
0.2000   -0.0667    0.1333
```

```
>> d = rank(a) % for rank
```

```
d =
```

```
3
```

```
>> d = size(a) % for finding size of a
```

```
d =
```

```
3    3
```

```
>> z = zeros(2,3) % zero matrix
```

```
z =
```

```
0    0    0  
0    0    0
```

```
>> y = ones(3,2) % one matrix
```

```
y =
```

```
1    1  
1    1  
1    1
```

```
>> x = eye(2,3) % unit matrix
```

```
x =
```

```
1    0    0  
0    1    0
```

```
>> x = eye(3,3) % unit matrix
```

```
x =
```

```
1    0    0  
0    1    0  
0    0    1
```

Vector

A vector is a quantity that has both magnitude and direction. It represents physical quantities like force, velocity, and displacement. Vectors are used in mathematics, physics, engineering, and computer graphics to model motion, forces, and directions. They can be represented in 2D or 3D as coordinates or arrows.

Some mathematical operations of vector are given below.

```
>> a = [1 2 3; 5 6 2; 1 0 4]
```

```
a =
```

```
1    2    3
5    6    2
1    0    4
```

```
>> b = [2 7 4; 3 0 1; 3 2 9]
```

```
b =
```

```
2    7    4
3    0    1
3    2    9
```

```
>> a + b % additon matrix a and b
```

```
ans =
```

```
3    9    7
8    6    3
4    2   13
```

```
>> a - b % subtraction matrix a and b
```

```
ans =
```

```
-1   -5   -1
2     6    1
-2   -2   -5
```

```
>> a * b % multiplication matrix a and b
```

```
ans =
```

```
17   13   33
34   39   44
14   15   40
```

```
>> a.*b % Element-wise multiplication
```

```
ans =
```

```
2    14   12
15    0    2
3     0   36
```

```
>> u = b / norm(b) % Unit Vector of b
```

```
u =
```

```
0.1678    0.5874    0.3357
0.2518         0    0.0839
0.2518    0.1678    0.7553
```

```
>> dp = dot(a,b) % Dot Product
```

```
dp =
```

```
20    14    50
```

```
>> cp = cross(a,b) % cross product
```

```
cp =
```

```
12    12    14
-1     -4   -11
-7   -42    -5
```

```
>> m = norm(a) % Magnitude of vector a
```

```
m =
```

```
8.8593
```

```
>> u = v1 / norm(a) % Unit Vector of a
Unrecognized function or variable 'v1'.
```

```
Did you mean:
```

```
>> u = a / norm(a) % Unit Vector of a
```

```
u =
```

```
0.1129    0.2258    0.3386
0.5644    0.6773    0.2258
0.1129         0    0.4515
```

```
>> a' % transpose of a vector
```

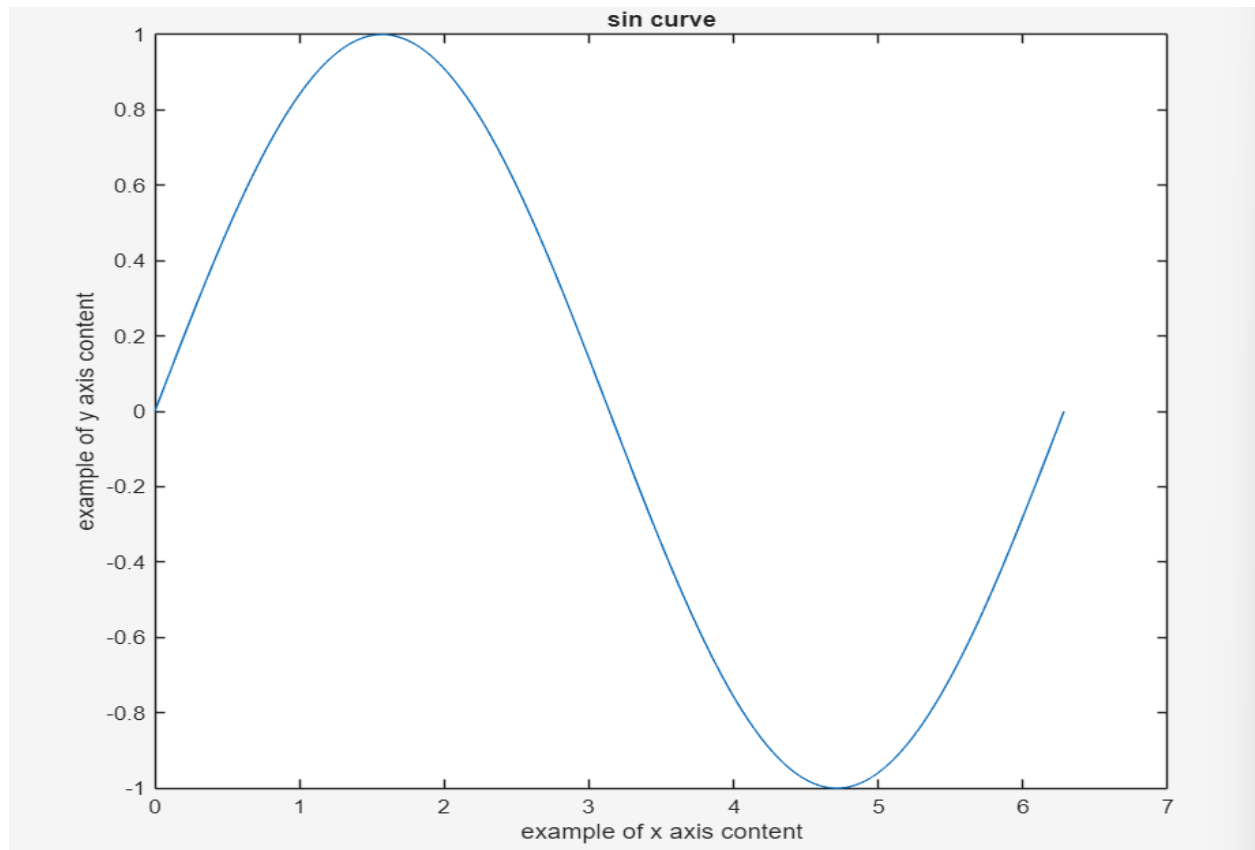
```
ans =
```

```
1    5    1
2    6    0
3    2    4
```

Sin Curve

```
>> x = 0:0.001:2*pi; % x values
y = sin(x);           % y values

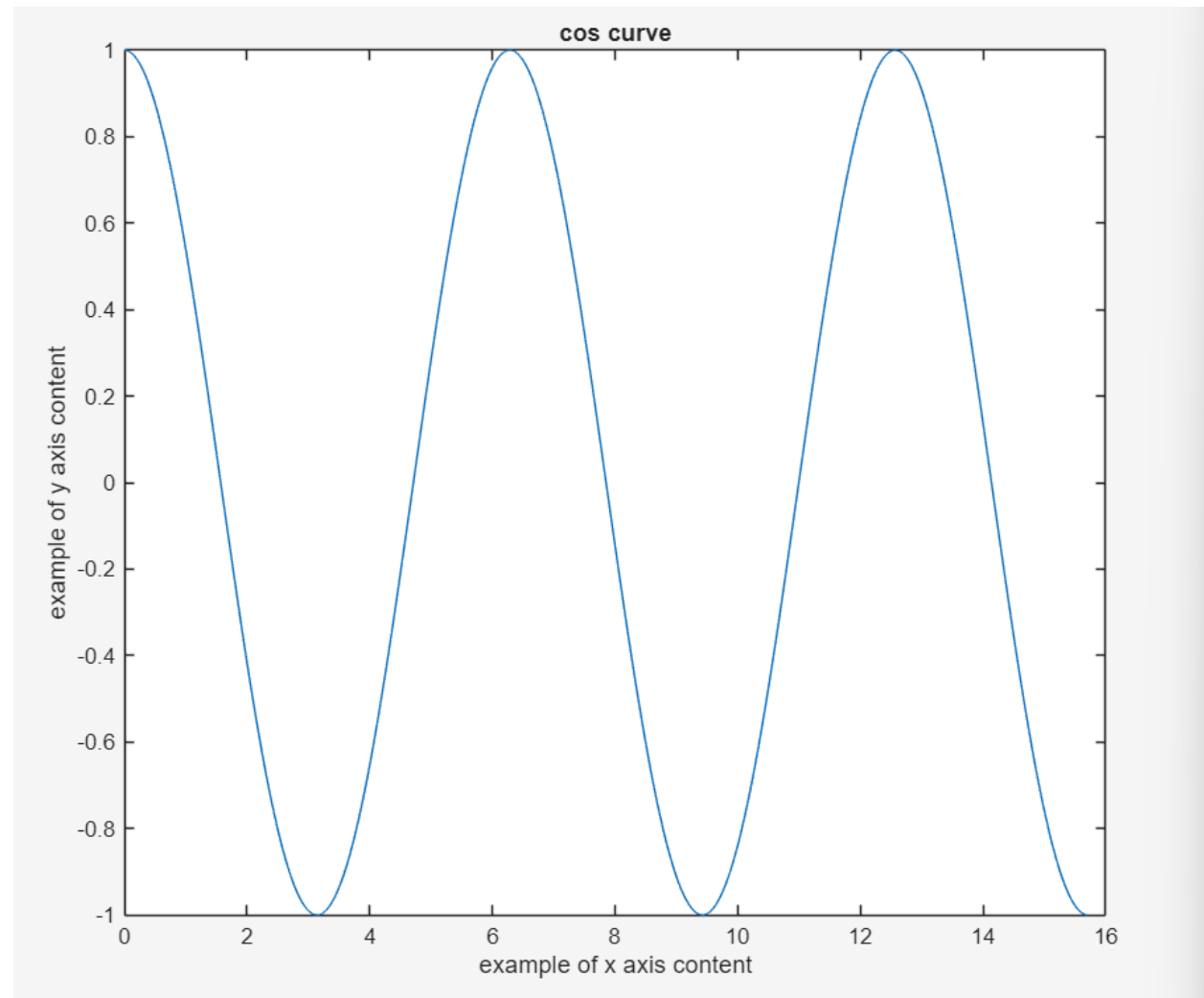
plot(x, y)
xlabel("example of x axis content")
ylabel("example of y axis content")
title("sin curve")
>> |
```



Cos curve

```
>> x = 0:0.001:5*pi; % x values
y = cos(x);          % y values

plot(x, y)
xlabel("example of x axis content")
ylabel("example of y axis content")
title("cos curve")
```



linear equation

A linear equation is an algebraic equation in which each variable appears only to the first power and is not multiplied by another variable. It represents a straight line in 2D, a plane in 3D, or a hyperplane in higher dimensions. Linear equations are the foundation of linear algebra and are widely used in mathematics, physics, engineering, and computer science for modeling relationships between quantities. They can have one or more variables, and their solutions are points or sets of points that satisfy the equation

```
>> % 2x + 132y = 189
>> % 8x + 12y = 120
>>
>> a = [3 132;8 12]

a =

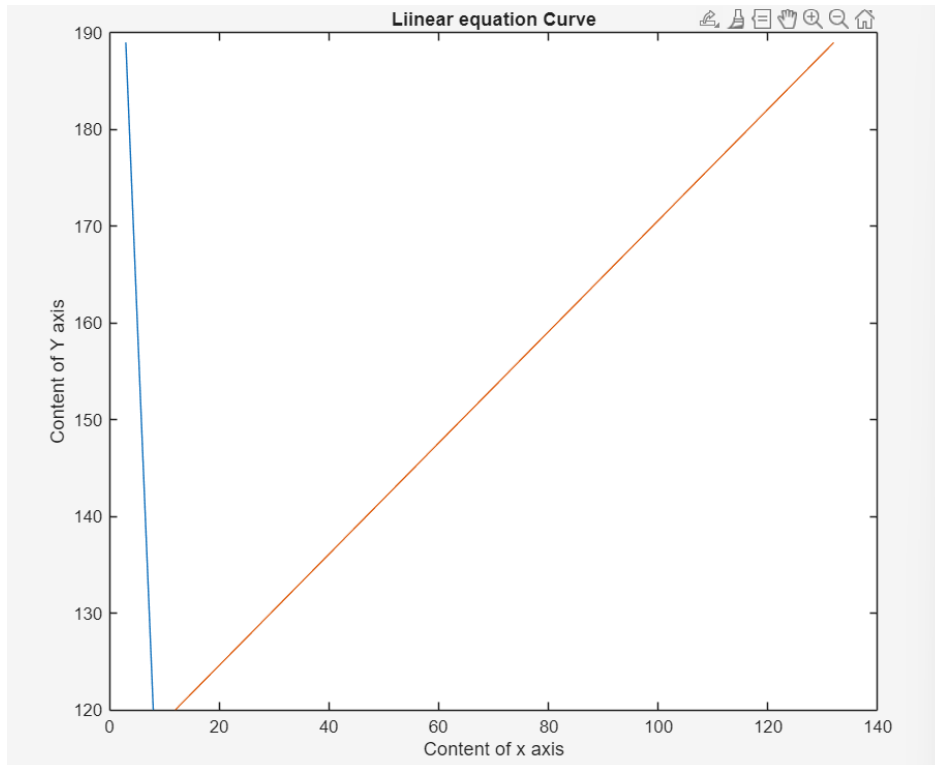
     3    132
     8     12

>> b = [189;120]

b =

    189
    120

>> plot(a,b)
>> xlabel("Content of x axis")
>> ylabel("Content of Y axis")
>> title("Linear equation Curve")
>> |
```



Quadratic Equation

A quadratic equation is an equation where the highest power of the variable is **2**, in the form $ax^2+bx+c=0$. It is used to model real-world problems, find maximum or minimum values, solve geometric or physics problems, and forms the basis for many applications in mathematics, engineering, and science.

Example of MATLAB:

```
>> % x^2-4x+3=0
>> x = [0:0.01:5]
>> plot(x,y)
>> xlabel("Content of x axis")
>> ylabel("Content of Y axis")
```

